

OFFICIAL ARCHITECTURE & CONFIGURATION MANUAL

Autogram Master Configuration Guide

Zero-Cost Autonomous Instagram Content Generation, Scheduling, Monetization & Publishing

Platform: Autogram AI v2.0 **Operating Cost:** \$0.00 / mo **Status:** Production Ready **Last Updated:** September 2026

Autogram Master Configuration & Operations Guide

The Complete Technical & Operational Manual for Zero-Cost Autonomous Instagram Content Generation, Scheduling, Monetization & Publishing

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1. Executive Architecture Overview

Autogram is an enterprise-grade, autonomous content production engine engineered to run at **\$0.00 operational cost**. The system operates across 5 decoupled layers:

```
[Layer A: Research & Scoring]
| • 14 Curated High-Signal Sources (Tech, AI, Growth, Dev)
| • 6-Pillar Diversity Scorer (+15.0 Diversity Bonus, -10.0 Repetition Penalty)
▼
[Layer B: Synthesis & Quality Gate]
| • Carousel Scriptwriting (8-10 slides, 1080x1350)
| • Fact-Checking against original source citations
| • 100-Point Quality Gate (Structure, typography, engagement)
| • Reels 30s-45s companion script & hashtag caption
▼
[Layer C: Visual AI & Rendering Engine]
| • FLUX.1 High-Resolution Hero Visual Generation (1080x1080)
| • Playwright Headless Chromium HTML/CSS slide rendering
| • Strict brand design system enforcement (inter typography, contrast ratios)
▼
[Layer D: Resilient Asset Hosting & Tunneling]
| • Local HTTP Server with path normalizer & CORS support (`free_host.py`)
| • Free Tunneling proxy (Localtunnel / Cloudflared / Ngrok)
▼
[Layer E: Meta Graph API Publishing]
| • Token health inspection & automatic 60-day refresh
| • Media container generation (item containers -> carousel container)
| • Publishing status polling & manifest logging
```

2. Environment Credentials & Secrets Configuration (.env)

Create your .env file in the root workspace directory (d:\Autogram\.env). Use this complete parameter reference:

```
# =====
# AUTOGRAM MASTER CONFIGURATION
# =====

# -----
# 1. OWNER VIP ACCESS & SECURITY
# -----
# Secret master passcode granting 100% free lifetime access to CLI & Web Studio
AUTOGRAM_OWNER_KEY=autogram_owner_vip_2026

# Secret key used for signing offline-verifiable HMAC-SHA256 client licenses
LICENSE_SECRET_KEY=autogram_hmac_master_secret_2026_x9k2p

# -----
# 2. FREE AI PROVIDER CREDENTIALS
# -----
# Google Gemini API Key (Free tier: 15 requests/minute, 1M tokens/minute)
# Get your free key at: https://aistudio.google.com/
GEMINI_API_KEY=your_gemini_api_key_here

# Groq Cloud API Key (Free tier: ultra-fast Llama 3.3 70B & 8B inference)
# Get your free key at: https://console.groq.com/
GROQ_API_KEY=your_groq_api_key_here

# Active LLM Provider Selection: "auto" | "gemini" | "groq" | "free_engine"
LLM_PROVIDER=auto

# Image Generation Provider: "pollinations" (FLUX.1, $0 cost, no key required)
IMAGE_PROVIDER=pollinations

# -----
# 3. META GRAPH API (INSTAGRAM PROFESSIONAL / BUSINESS)
# -----
# Instagram Business Account ID (Numeric ID, e.g. 17841400000000000)
INSTAGRAM_ACCOUNT_ID=your_instagram_account_id_here

# Meta App ID & Secret (from https://developers.facebook.com/)
META_APP_ID=your_facebook_app_id
META_APP_SECRET=your_facebook_app_secret

# Long-Lived Page/User Access Token (60 days validity)
META_ACCESS_TOKEN=your_long_lived_meta_access_token_here

# -----
# 4. ASSET HOSTING & TUNNELING
# -----
# Asset storage mode: "free_host" (default) | "r2" | "supabase" | "local"
STORAGE_PROVIDER=free_host

# Local asset server host & port
LOCAL_HOST_PORT=8000
LOCAL_HOST_DIR=output

# Public Tunnel URL (Updated automatically by tunneling scripts)
# Example: https://quiet-fox-92.loca.lt or https://autogram.serveo.net
PUBLIC_ASSET_BASE_URL=http://localhost:8000

# -----
# 5. SCHEDULING & TIMEZONE
# -----
```

```
# Daily autonomous posting schedule (24-hour format: HH:MM)
POSTING_TIME=19:30
TIMEZONE=Asia/Kolkata

# -----
# 6. MONETIZATION & BILLING
# -----
UPI_ID=6202442690@jio
WHATSAPP_NUMBER=+916202442690
```

3. Zero-Cost AI Provider Setup

Autogram guarantees uninterrupted autonomous operation through a 4-tier cascading intelligence model:

3.1 Built-in Autonomous Knowledge Engine (\$0.00, Zero External Keys)

- **Engine File:** `src/content/free_knowledge_engine.py`
- If external API keys are missing, exhausted, or return rate limits (429), the Built-in Engine automatically synthesizes research into structured 8-slide decks, fact-checks content, generates 35s Reels scripts, and formats hashtag captions.
- Produces 100% Quality Gate pass scores with zero internet API dependency.

3.2 Google Gemini 1.5 / 2.0 Flash (Free Tier)

1. Navigate to [Google AI Studio](#).
2. Log in with your Google account and click "**Create API Key**".
3. Copy the key and set `GEMINI_API_KEY=AIZA...` in your `.env`.
4. **Limits:** 15 Requests Per Minute (RPM), 1,500 Requests Per Day (RPD), 1,000,000 tokens/min. Autogram requires only ~3 requests per daily run, consuming less than 0.2% of daily allowances.

3.3 Groq Cloud API (Free Tier)

1. Visit [Groq Cloud Console](#).
2. Create an account and generate an API key in **API Keys**.
3. Set `GROQ_API_KEY=gsk...` in your `.env`.
4. Provides high-speed Llama 3.3 70B Versatile inference for fact-checking and scriptwriting at ~300 tokens/second.

3.4 Pollinations AI FLUX.1 (Free Visual Generation)

- **Engine File:** `src/content/image_generator.py`
- Generates photorealistic and graphic hero visuals via the state-of-the-art **FLUX.1** diffusion model.
- Requires **no API key** and has **no token fees**.
- Autogram formats prompts with architectural aesthetics: `minimalist modern aesthetic, clean studio lighting, high contrast, 8k, cinematic framing`.

4. Meta Graph API & Instagram Professional Setup

Meta requires an **Instagram Professional (Creator or Business) Account** connected to a **Facebook Page**.

4.1 Account Prerequisites

1. Open Instagram on your mobile phone: **Settings** → **Account** → **Switch to Professional Account**. Select "**Creator**" or "**Business**".
2. Open Facebook and create or open your Business/Creator Page.
3. Link the two: On your Facebook Page, navigate to **Settings** → **Linked Accounts** → **Instagram** → **Connect Account**.

4.2 Meta Developer App Creation

1. Go to [Meta for Developers](#) and click **My Apps** → **Create App**.
2. Select "**Other**" → App Type: "**Business**".
3. Under **Add products to your app**, add **Instagram Graph API**.
4. Navigate to **App Settings** → **Basic** to retrieve:
 - `META_APP_ID` (App ID)
 - `META_APP_SECRET` (App Secret)

4.3 Required Permissions

Generate a User Token in the **Graph API Explorer** with these permissions:

- `instagram_basic`
- `instagram_content_publish`
- `pages_read_engagement`
- `pages_show_list`
- `public_profile`

4.4 Converting to a 60-Day Long-Lived Token

Run this conversion request in your browser or terminal to extend your temporary token to 60 days:

```
curl -X GET "https://graph.facebook.com/v23.0/oauth/access_token?
grant_type=fb_exchange_token&
client_id=YOUR_META_APP_ID&
client_secret=YOUR_META_APP_SECRET&
fb_exchange_token=YOUR_SHORT_LIVED_TOKEN"
```

Copy the returned `access_token` and paste it into `.env` as `META_ACCESS_TOKEN`.

4.5 Finding Your Instagram Account ID

Run:

```
curl -X GET "https://graph.facebook.com/v23.0/me/accounts?access_token=YOUR_META_ACCESS_TOKEN"
```

Find your page, take its `id`, then query:

```
curl -X GET "https://graph.facebook.com/v23.0/PAGE_ID?fields=instagram_business_account&access_token=YOUR_META_TOKEN"
```

The numeric ID returned under `instagram_business_account.id` is your `INSTAGRAM_ACCOUNT_ID`.

5. Content Strategy, Pillars & Knowledge Feeds

5.1 The 6 Core Content Pillars

Autogram cycles through 6 high-converting B2B content categories:

PILLAR	FOCUS & ARCHETYPE	PRIMARY SLIDE TEMPLATE
1. AI Tool Breakdown	Practical review, workflow demo, prompt template	Comparison, Checklist
2. Prompting & Workflow	Actionable operating system, step-by-step framework	Framework, Standard
3. Marketing Psychology	Behavioral triggers, conversion laws, copywriting	Hook, Diagram, Takeaway
4. Tech Industry Explainer	Architecture, open-source releases, API teardowns	Diagram, Standard
5. Career & Skills	Market compensation, high-leverage skills, positioning	Checklist, Framework
6. Myth-Bust / Contrarian	Challenging industry dogmas with empirical data	Comparison, Takeaway

5.2 Curated Knowledge Feeds (`data/seed_sources.json`)

The source catalog contains 14 curated high-signal feeds:

- **AI Tool Feeds:** GitHub Trending AI, Product Hunt Daily, HuggingFace Papers
- **Engineering Feeds:** MIT Tech Review AI, Hacker News Top, TechCrunch AI
- **Growth & Marketing:** Marketing Examined, Growth Design Teardowns, Demand Curve
- **Developer Ecosystem:** Meta for Developers, OpenAI Research, PyTorch Updates

To add a new custom feed, append an entry to `data/seed_sources.json`:

```
{
  "name": "ArXiv AI Research",
  "url": "https://export.arxiv.org/rss/cs.AI",
  "pillar": "Tech Industry Explainer",
  "type": "rss",
  "authority_weight": 1.2
}
```

6. Slide Rendering Engine & Visual Design System

Autogram renders publication-quality 1080×1350 JPEG slides using headless Chromium via Playwright, ensuring crisp typography and no blurry diffusion artifacts.

```
renderer/
├── css/
│   └── design-system.css      # Core typography, dark-mode colors, glassmorphism tokens
├── templates/
│   ├── hook.html             # Slide 1: High-impact title, badge, reading time, author
│   ├── standard.html         # Core content slides with headline, key points, highlight cards
│   ├── comparison.html       # Side-by-side vs comparison layout (Good vs Bad, Old vs New)
│   ├── framework.html        # Numbered sequential steps with progress indicators
│   ├── checklist.html        # Action items with styled checkboxes
│   ├── diagram.html          # Visual concept architecture layout
│   ├── takeaway.html         # High-impact summary cards
│   └── cta.html              # Final slide: Save, Share, Follow, Free Resource CTA
└── render.py                 # Playwright automation script (device scale factor: 1.0)
```

Design Standards Enforced

- **Aspect Ratio:** 4:5 Portrait (`1080px × 1350px`)
- **Safe Zone:** 100px padding around borders (safe from Instagram UI buttons)
- **Palette:** Dark Mode Obsidian (`#090A0F`), Electric Indigo (`#6366F1`), Cyan Glow (`#06B6D4`), Emerald Accent (`#10B981`)
- **Typography:** Inter / Plus Jakarta Sans with strict typographic scale (`clamp(32px, 4vw, 56px)`)

7. Asset Hosting & Free Tunneling Infrastructure

Meta's Graph API requires publicly accessible HTTPS URLs to download carousel images. Autogram offers a zero-cost local hosting solution:

```
sequenceDiagram
    participant O as Autogram Orchestrator
    participant H as free_host.py (Port 8000)
    participant T as Public Tunnel (Localtunnel/Cloudflared)
    participant M as Meta Graph API

    O->>H: Start HTTP server serving output/ directory
    O->>T: Open public tunnel to http://localhost:8000
    T-->>O: Public URL: https://autogram-live.local.lt
    O->>M: POST /media?image_url=https://autogram-live.local.lt/output/2026-09-09/slide_01.jpg
    M->>T: Fetch slide_01.jpg
    T->>H: Forward GET /output/2026-09-09/slide_01.jpg
    H-->>M: Return 1080x1350 JPEG (HTTP 200)
    M-->>O: Container ID created successfully
```

7.1 Running the Asset Server

Start the local server manually:

```
python free_host.py
```

- Listens on `http://0.0.0.0:8000`

- Normalizes URL paths automatically (strips accidental `/output/` duplicates)
- Includes CORS headers and `/health` probe

7.2 Starting a Free Public Tunnel

In a second terminal, launch a tunnel pointing to port 8000:

Option A: Localtunnel (No Account Required)

```
npx localtunnel --port 8000 --subdomain autogram-feed
```

Then set `PUBLIC_ASSET_BASE_URL=https://autogram-feed.localt` in `.env`.

Option B: Cloudflare Quick Tunnel (Free, Fast, Secure)

```
cloudflared tunnel --url http://localhost:8000
```

8. Access Control, Paywall & Client Monetization

Autogram includes an enterprise cryptographic licensing and monetization system:

```
graph TD
    User[User / Client] --> Check{Identity Check}
    Check -->|Owner Key Detected| VIP[👑 Owner VIP Status]
    VIP --> Unlocked[100% Free Lifetime Access to CLI & Web Studio]

    Check -->|Client License Key| Client[📁 Client Workspace]
    Client --> PaidUse[Gated to Client Tier: Starter / Growth / Enterprise]

    Check -->|No Key / Public Visitor| Paywall[🔒 Payment Gate]
    Paywall --> UPI[UPI Payment: 6202442690@jio]
    Paywall --> QR[Scan Dynamic UPI QR Code]
    UPI --> WA[Send Screenshot on WhatsApp: +91 6202442690]
    WA --> OwnerIssues[Owner Issues Signed Key: AG-GROWTH-XXXX]
    OwnerIssues --> ClientEnters[Client Unlocks Account]
```

8.1 Owner Free VIP Access

- In `.env`: `AUTOGRAM_OWNER_KEY=autogram_owner_vip_2026`
- **Instant Web Unlock:** Visit `https://autogram-ai.vercel.app/?admin=autogram_owner_vip_2026` (permanently saves VIP status to `localStorage`).
- **CLI Unlock:** Automatic on every terminal run.

8.2 Client Pricing & Payment Setup

- **UPI ID:** `6202442690@jio`
- **WhatsApp Support:** `+91 6202442690`

- **Plans:**
 1. **Starter Autopilot:** ₹39,999 / mo (\$497 USD) — 3 Carousels/week
 2. **Growth Autopilot:** ₹79,999 / mo (\$997 USD) — Daily Carousel + Reels Scripts
 3. **Enterprise Swarm:** ₹1,99,999 / mo (\$2,497 USD) — Custom Pillars, Multi-Account

8.3 Issuing a Client License Key

When a customer pays via UPI and sends their screenshot to WhatsApp, issue their key using either method:

Method 1: From the Terminal

```
python -m src.auth.licensing --issue --client "Acme Corp" --tier growth --days 30
```

Output: AG-GROWTH-68A1-ACMECORP-8F3C

Method 2: In the Web Browser

1. Open the Web Studio in Owner Mode.
2. Under 🗑️ **Owner License Dispenser**, enter the client's name and select tier.
3. Click **"Issue Key"** and copy the resulting string to WhatsApp.

9. Operational Execution Modes & CLI Reference

COMMAND	PURPOSE
python orchestrator.py --run-all --dry-run	Full Dry Run: Tests all 5 layers, writes slides and scripts to output/ , simulates Meta API without publishing.
python orchestrator.py --run-all	Live Production Publish: Runs pipeline and publishes carousel to live Instagram account.
python orchestrator.py --schedule	24/7 Autonomous Daemon: Keeps running locally, triggering the pipeline automatically at POSTING_TIME .
python orchestrator.py --test	Health Diagnostics: Validates license, verifies templates, and checks database connections.
python -m src.content.image_generator --prompt "AI Brain"	Visual Generator: Generates standalone 1080x1080 FLUX.1 hero image.
python renderer/render.py	Renderer Test: Compiles sample slides to verify Playwright Chromium engine.
python -m pytest tests -v	Full Test Suite: Runs all 16 unit and integration tests.

10. Cloud Serverless Automation (GitHub Actions)

Autogram can run completely serverless in GitHub Actions with zero hosting costs.

The workflow is configured in `.github/workflows/daily-post.yml`:

- **Schedule:** Triggers daily at `14:00 UTC` (`19:30 IST`).
 - **Dependencies:** Sets up Python 3.12, installs Playwright Chromium dependencies, and executes `python orchestrator.py --run-all`.
 - **Required GitHub Secrets** (under **Settings** → **Secrets and variables** → **Actions**):
 - `AUTOGRAM_OWNER_KEY`
 - `GEMINI_API_KEY`
 - `GROQ_API_KEY`
 - `INSTAGRAM_ACCOUNT_ID`
 - `META_ACCESS_TOKEN`
-

11. Web Application & Vercel Production Deployment

The landing page and Web Studio are hosted on Vercel:

- **Live URL:** `https://autogram-ai.vercel.app`
- **Configuration:** `vercel.json`

To deploy updates to Vercel:

```
npx vercel --prod --yes
```

12. Verification, Diagnostics & Troubleshooting

Common Diagnostics

ISSUE	ROOT CAUSE	SOLUTION
Meta Graph API 400 (Invalid Image URL)	Meta's crawler cannot reach the asset URL.	Verify your tunnel is active (<code>curl -I YOUR_TUNNEL_URL/health</code>). Ensure <code>PUBLIC_ASSET_BASE_URL</code> in <code>.env</code> matches the active tunnel.
Meta Graph API 190 (Token Expired)	The 60-day token has expired.	Re-run the token exchange script in Section 4.4 and update <code>META_ACCESS_TOKEN</code> .
Quality Gate Score < 80	Content draft had excessive jargon or missing hook.	The system automatically retries with a fallback prompt. If persistent, verify <code>src/content/quality_gate.py</code> .
Playwright Browser Missing	Chromium binaries not downloaded.	Run <code>playwright install chromium</code> .
Client Key Invalid / Tampered	HMAC signature mismatch or expired timestamp.	Ensure <code>LICENSE_SECRET_KEY</code> on server matches the key used when issuing the license.

Verification Checklist

Before enabling live unattended daily posting, ensure:

- [x] All 16 automated tests pass (`python -m pytest tests -v`).
- [x] A dry-run completes with a 100/100 Quality Gate score (`python orchestrator.py --run-all --dry-run`).
- [x] Owner VIP key is verified in `.env` .
- [x] Meta Access Token has at least 14 days of validity remaining.
- [x] Public asset hosting server is reachable over HTTPS.